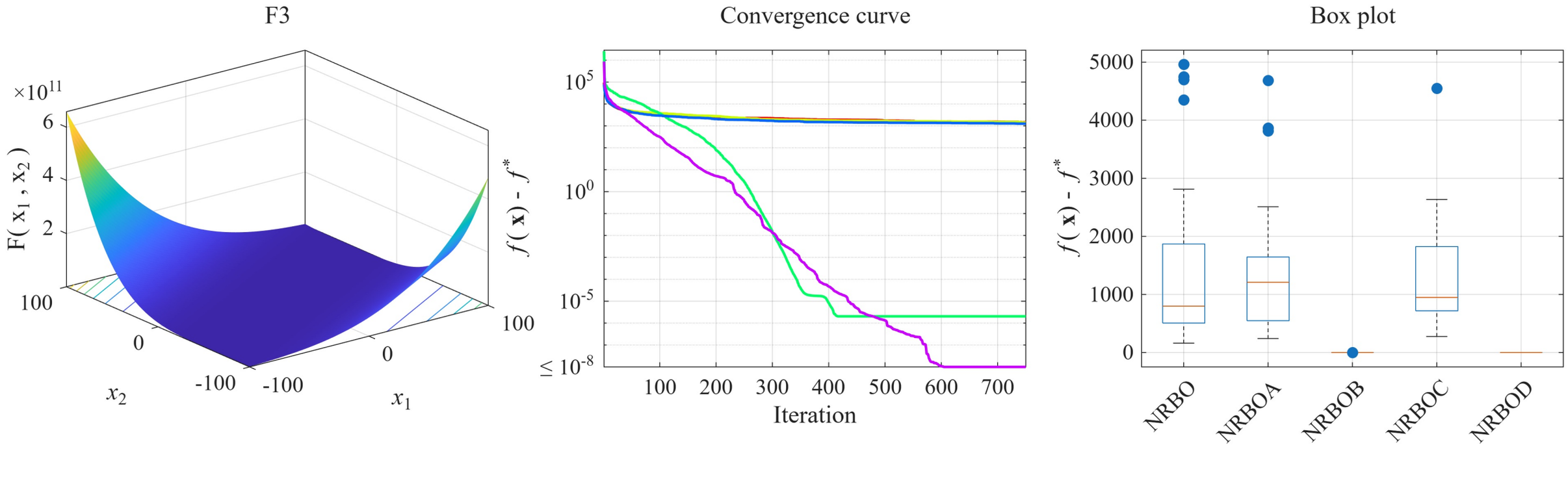

F3
×10^11
6
4
2
F(x1 , x2)
100
0
-100
x2
-100
0
x1
Convergence curve
10^5
10^0
10^-5
≤ 10^-8
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
5000
4000
3000
2000
1000
0
f(x) - f*
NRBO
NRBOA
NRBOB
NRBOC
NRBOD
NRBO
NRBOA
NRBOB
NRBOC
NRBOD